MATHEUS RENNAN

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SUMMARY

Software developer with experience in full-stack development, AI, and software architecture. Skilled in building scalable and efficient applications using .NET Core. Experienced in API integration and system performance optimization. Proven ability to contribute to high-impact projects, including the development of artificial intelligence solutions with experience in LLMs, RAG, Computer Vision, and innovative AI applications.

EDUCATION

Bachelor of Computer Science, Federal University of Cariri (UFCA) Sep 2022 - Present
Research scholarship recipient.
Undergraduate Researcher since Feb 2025, working on the development of LLMs for legal interpretation and analysis, with a focus on data collection, model training, fine-tuning, evaluation, and prompt engineering.

Bachelor of Civil Engineering, Federal University of Cariri (UFCA) Aug 2018 - Aug 2022
Teaching Assistant for Calculus I and II.

SKILLS

Languages	C#, Java, Python, JavaScript, SQL, HTML/CSS
Frameworks	.NET 10, .NET 8, .NET 6+, ASP.NET Core, Spring Boot, Flask, FastAPI
AI / Data	LangChain, LlamaIndex, RAG, Vector Embeddings, OpenAI API, Hugging Face, RAGAS
Cloud / DevOps	Azure, Azure DevOps, AWS, Docker, Kubernetes, Git
Messaging	Azure Service Bus, SQS (Concepts of RabbitMQ and Kafka)
Databases	SQL Server, PostgreSQL, Entity Framework Core, Dapper, Spring Data JPA
Architecture	DDD, CQRS, Hexagonal Architecture, Layered Architecture, Microservices, RESTful APIs
Practices	Clean Architecture, SOLID, Design Patterns, Clean Code, Scrum, Kanban, Gitflow
Languages	English (C1 Advanced), Portuguese (Native)

EXPERIENCE

Full Stack Software Engineer Apr 2026 – Present
LUXCLUSIF / FARFETCH – Remote

- Developing and maintaining cloud-based microservices using .NET 8/10 and Microsoft Azure.
- Implementing and maintaining RESTful API security using authentication and authorization with JWT and OAuth2/OIDC.
- Working with Azure Service Bus for asynchronous messaging and Azure App Service for application deployment.
- Monitoring applications with Application Insights and implementing feature flags for controlled releases.
- Automating infrastructure with Terraform and working with Scrum/Kanban and Gitflow-based version control in Bitbucket and Jira.
- Contributing to bug fixes, new feature development, and the continuous evolution of the platform.

Generative AI Researcher (FINEP) Feb 2025 – Present
Federal University of Cariri (UFCA) – Remote

- Working on the project "Pre-training and Fine-tuning LLMs for Legal Interpretation and Analysis," focused on improving automated legal reasoning for Portuguese legal documents.
- Generating, cleaning, and optimizing datasets for pre-training and fine-tuning stages, including domain-specific corpora for classifiers and task-oriented LLMs.

- Developing prompt engineering strategies to improve relevance, accuracy, and extraction quality in legal-domain LLM applications.
- Building Retrieval-Augmented Generation (RAG) pipelines with LangChain and LlamaIndex to improve the retrieval of case law and legal documents from natural language queries.
- Integrating external structured and unstructured data sources to improve legal information extraction and automated analysis workflows.
- Training custom tokenizers and performing continuous pre-training and layer-level adjustments using Hugging Face tools for legal-domain specialization.
- Evaluating LLM pipelines with Databricks and RAGAS using metrics such as response relevance and generation quality.

Mid-Level Software Engineer Dec 2024 – Apr 2026
Avenue Code – Remote

- Applying Domain-Driven Design (DDD), Command Query Responsibility Segregation (CQRS) using MediatR and AutoMapper, Bounded Context, Ubiquitous Language, and Hexagonal Architecture to ensure clear separation of responsibilities and long-term adaptability.
- Implementing orchestration flows in BFF/BFI layers, centralizing business rules and improving performance and maintainability.
- Developing backend services with ASP.NET Core applying SOLID principles and Design Patterns, optimizing SQL queries, and ensuring scalable data access with Entity Framework Core and Dapper.
- Writing unit and integration tests with xUnit, NSubstitute, and Faker (Bogus), aiming for high coverage of critical business rules.
- Contributing to the modernization of a large-scale digital banking platform, with a strong focus on user experience, reliability, and security.
- Using Docker and Kubernetes for containerization and orchestration in cloud-native environments.
- Working in an Azure-based environment with Azure DevOps for CI/CD pipelines, version control, and project tracking.

Software Developer Jul 2023 – Nov 2024
BRISA Sociedade para o Desenvolvimento da Tecnologia da Informação – Remote

- Built and optimized Web APIs using ASP.NET Core with Layered Architecture, implementing CRUD operations with Entity Framework Core and PostgreSQL.
- Ensured API performance and reliability through unit testing and consistent integration with frontend systems.
- Developed a document validation microservice using AWS Lambda, triggered by Amazon SQS and integrated with Amazon Textract and OpenAI LLMs through Semantic Kernel in .NET.
- Automated OCR and contextual data extraction workflows, reducing manual effort and increasing operational efficiency.
- Contributed to the backend of an internal management platform using Spring Boot, applying DDD, dependency injection, and layered architecture.
- Implemented RESTful APIs with Spring Data JPA and PostgreSQL, ensuring security with Spring Security and role-based access control.
- Designed and developed intelligent chatbots using Python, Django, and Flask, integrating LLMs to provide contextual responses to users and students.
- Implemented RAG pipelines with LangChain, vector embeddings, and semantic retrieval, as well as speech-to-text capabilities using OpenAI Whisper.
- Applied prompt engineering, context caching, and embedding-based retrieval to improve conversational performance and response quality.

Artificial Intelligence Intern Jan 2023 – Aug 2023
n.0 Tecnologia – Remote

- Developed deep learning models for equine behavior detection using YOLO, YOLOX, Detectron2, and OpenCV, applying object detection and instance segmentation techniques.
- Created algorithms to interpret neural network outputs and generate actionable insights for image and video analysis.
- Developed a natural language to SQL translation system using LLMs, mapping user intent to structured queries.
- Supported the integration and deployment of Generative AI solutions in real-world scenarios using LangChain, Ollama, and Flask.

CERTIFICATES

Databricks Certified Generative AI Engineer Associate Jan 2026
Databricks
Credential ID: 172617930 Expires Jan 2028

AI-Assisted Certified Professional Feb 2025
AI/R

Applying AI Principles with Google Cloud Jan 2025
Google
Skills: Generative AI, Artificial Intelligence

EF SET English Certificate – 67/100 (C1 Advanced) Apr 2024
EF SET